

Zelestra Hackathon: Phase 2

Assumptions:

- All SCB in each inverter zone generate same amt of current
- Ideal conditions considered to be STC : 25 °C
- Assumed temperature coefficient of 0.45%
- Humidity and temperature contribute equally (not physically accurate).
- Temperature scaled arbitrarily by 100.

Methodology

(Assume conversion of units wherever necessary)

Angle loss

Loss in each zone due to difference in Tracer angle objective and current angle.

1. Difference between objective and actual angle of tracker
2. Find cosine loss using $(1 - \cos(\text{difference in angle}))$
3. Multiply cosine loss with Global horizontal irradiance for the zone resulting in Irradiance loss due to angle.
4. Multiply irradiance loss with inverter capacity and inverter efficiency in that time interval, resulting in Power loss ,due to angle difference in MW.

Soiling loss

Loss in each zone due to soiling. It is found using cell level irradiance values (from `celulas_ctin0*_cc_0*_*_ir_cel_*` features) and global irradiance values.

$$\begin{aligned} & \text{SL} = \text{SOILING LOSS} \\ & \text{RP} = \text{RATED POWER} = 3800 \\ & \frac{\left(\frac{(\text{LOCAL_IRRADIANCE} - \text{GLOBAL_IRRADIANCE})}{1000} \right) \times (\text{RP} \times \text{INVERTER EFFICIENCY})}{1000} \\ & \text{SOILING LOSS} = \text{SL} - \text{SHADING_LOSS} \end{aligned}$$

1. Find average of cell level irradiance values from each zone and store in separate features of dataframe.
2. Find zone level expected irradiance – sum of GII and Reflected GII.
3. Find difference between the average and zone level expected irradiance.
4. Find product of the zone's inverter capacity , efficiency and found difference.
5. The resulting value is power loss due to it in MW.

Temperature loss

Loss in each zone due to Module Temperature . It is found using cell level Temperature values (from `celulas_ctin0*_cc_0*_t_mod` features) .

1. Find average of module temperature in each zone.
2. Use standard formula to find the loss due to temperature

$$P_{\text{loss}} = P_{\text{rated}} \times \gamma \times (T_{\text{cell}} - 25), \gamma \text{ assumed to be } 0.45\%$$

3. P_{rated} is the inverter capacity of each zone, Resulting in kW later converted to loss in MWs.

Cloud cover loss

Loss in cloud cover temperature and

Inverter

Inverter converting DC input

TL = TEMPERATURE LOSS
RP = RATED POWER = 3800
T_MOD = ACTUAL CELL TEMPERATURE

$$TL = \frac{(RP \times (-0.0045)) \times (T_MOD - 25)}{1000}$$

HR = RELATIVE HUMIDITY
TR = TEMPERATURE

$$CC = \frac{HR/100 + TR/100}{2}$$

each zone due to estimated using humidity.

Efficiency

efficiency in to AC output.

String to Inverter

Assumption:
have similar power
string refers to a
connected to Zone 3

IE = INVERTER EFFICIENCY
I_DC = INVERTER INPUT IN DC
O_AC = INVERTER OUTPUT IN AC

SIPL = STRING_TO_INVERTER_POWER_LOSS
STR_A = COMBINED STRING OUTPUT IN AMPERE
STR_V = DC BOX COMBINED DC VOLTAGE

$$SIPL = \left(\frac{STR_A \times STR_V}{1E-6} \right) - INVERTER\ DC\ INPUT$$

Angle Loss

Loss due to

Power Loss

All strings in a zone
generation and here
SCB and 13 SCBs
and 12 to zone 8.

Tracker angle loss

AL = ANGLE_LOSS
SL = OBJECTIVE POSITION
TP = TARGET POSITION

$$AL = \left(\frac{1 - \cos(OBJ_P - TRG_P) \times GLOBAL_IRRADIANCE}{1E-6} \right) \times INVERTER\ EFFICIENCY$$

Shading Loss

Loss due to Shade

- Determine expected irradiance
For each time point, calculate the expected irradiance as 80% of the highest irradiance observed within a rolling window of recent values for that day.
- Identify shading conditions
Mark a shading event when the actual average cell irradiance is lower than the expected irradiance, provided the reference irradiance is above 100 (indicating sunlight is present).
- Calculate shading loss
Compute the difference between the expected and actual irradiance during shading events to quantify the loss in irradiance.
- Convert to megawatts
Translate the shading loss in irradiance into power loss (in megawatts) by factoring in the array size and efficiency.

Loss Attribution Sample from Loss Attribution Notebook

Quant Loss have Quantified values for each loss

quant_loss.head(20)

(14) ✓ 0.0s

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	datetime	Zone	INVERTER	String	String Input	Shading	CloudCover	Temperature Effect	Soiling	Inverter_efficiency_loss	String_to_inverter_Power_Loss	Power_loss_due_to_Angles
0	2024-10-01 08:00:00	CTIN 03	INV-3	8	1	0.0	0.532341	0.193800	0.002242	0.041168	0.000243	0.001248
1	2024-10-01 08:00:00	CTIN 03	INV-3	10	1	0.0	0.532341	0.193800	0.002242	0.041168	0.001051	0.001248
2	2024-10-01 08:00:00	CTIN 08	INV-8	9	1	0.0	0.538657	0.203234	0.003660	0.012059	0.000873	0.000103
3	2024-10-01 08:00:00	CTIN 08	INV-8	12	1	0.0	0.538657	0.203234	0.003660	0.012059	0.000597	0.000103
4	2024-10-01 08:15:00	CTIN 03	INV-3	8	1	0.0	0.526895	0.187459	0.057376	0.025967	0.001138	0.004849
5	2024-10-01 08:15:00	CTIN 03	INV-3	10	1	0.0	0.526895	0.187459	0.057376	0.025967	0.009359	0.004849
6	2024-10-01 08:15:00	CTIN 08	INV-8	9	1	0.0	0.534847	0.205200	0.057084	0.019669	0.002812	0.002067
7	2024-10-01 08:15:00	CTIN 08	INV-8	12	1	0.0	0.534847	0.205200	0.057084	0.019669	0.000171	0.002067
8	2024-10-01 08:30:00	CTIN 03	INV-3	8	1	0.0	0.525748	0.166236	0.112534	0.016760	0.002957	0.005154
9	2024-10-01 08:30:00	CTIN 03	INV-3	10	1	0.0	0.525748	0.166236	0.112534	0.016760	0.021278	0.005154
10	2024-10-01 08:30:00	CTIN 08	INV-8	9	1	0.0	0.531732	0.184965	0.114203	0.014948	0.005512	0.005068
11	2024-10-01 08:30:00	CTIN 08	INV-8	12	1	0.0	0.531732	0.184965	0.114203	0.014948	0.002882	0.005068
12	2024-10-01 08:45:00	CTIN 03	INV-3	8	1	0.0	0.521711	0.129120	0.176552	0.018387	0.008905	0.018486
13	2024-10-01 08:45:00	CTIN 03	INV-3	10	1	0.0	0.521711	0.129120	0.176552	0.018387	0.032730	0.018486
14	2024-10-01 08:45:00	CTIN 08	INV-8	9	1	0.0	0.515009	0.141579	0.129361	0.023502	0.002569	0.001950
15	2024-10-01 08:45:00	CTIN 08	INV-8	12	1	0.0	0.515009	0.141579	0.129361	0.023502	0.009403	0.001950
16	2024-10-01 09:00:00	CTIN 03	INV-3	8	1	0.0	0.513281	0.063418	0.173378	0.019375	0.011868	0.104203
17	2024-10-01 09:00:00	CTIN 03	INV-3	10	1	0.0	0.513281	0.063418	0.173378	0.019375	0.041094	0.104203
18	2024-10-01 09:00:00	CTIN 08	INV-8	9	1	0.0	0.499340	0.082414	0.089983	0.020316	0.001639	0.011288
19	2024-10-01 09:00:00	CTIN 08	INV-8	12	1	0.0	0.499340	0.082414	0.089983	0.020316	0.018853	0.011288

Binary Loss DF have boolean flags for each loss

binary_loss.head(20)

(15) ✓ 0.0s

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	datetime	Zone	INVERTER	String	String Input	Shading	CloudCover	Temperature Effect	Soiling	Inverter_efficiency_loss	String_to_inverter_Power_Loss	Power_loss_due_to_Angles
0	2024-10-01 08:00:00	CTIN 03	INV-3	8	1	0	1	1	0	1	0	0
1	2024-10-01 08:00:00	CTIN 03	INV-3	10	1	0	1	1	0	1	0	0
2	2024-10-01 08:00:00	CTIN 08	INV-8	9	1	0	1	1	0	0	0	0
3	2024-10-01 08:00:00	CTIN 08	INV-8	12	1	0	1	1	0	0	0	0
4	2024-10-01 08:15:00	CTIN 03	INV-3	8	1	0	1	1	0	0	0	0
5	2024-10-01 08:15:00	CTIN 03	INV-3	10	1	0	1	1	0	0	0	0
6	2024-10-01 08:15:00	CTIN 08	INV-8	9	1	0	1	1	0	0	0	0
7	2024-10-01 08:15:00	CTIN 08	INV-8	12	1	0	1	1	0	0	0	0
8	2024-10-01 08:30:00	CTIN 03	INV-3	8	1	0	0	1	0	0	0	0
9	2024-10-01 08:30:00	CTIN 03	INV-3	10	1	0	0	1	0	0	1	0
10	2024-10-01 08:30:00	CTIN 08	INV-8	9	1	0	1	1	0	0	0	0
11	2024-10-01 08:30:00	CTIN 08	INV-8	12	1	0	1	1	0	0	0	0
12	2024-10-01 08:45:00	CTIN 03	INV-3	8	1	0	0	1	1	0	0	0
13	2024-10-01 08:45:00	CTIN 03	INV-3	10	1	0	0	1	1	0	1	0
14	2024-10-01 08:45:00	CTIN 08	INV-8	9	1	0	0	1	1	1	0	0
15	2024-10-01 08:45:00	CTIN 08	INV-8	12	1	0	0	1	1	1	0	0
16	2024-10-01 09:00:00	CTIN 03	INV-3	8	1	0	0	0	1	0	1	0
17	2024-10-01 09:00:00	CTIN 03	INV-3	10	1	0	0	0	1	0	1	0
18	2024-10-01 09:00:00	CTIN 08	INV-8	9	1	0	0	0	0	0	0	0
19	2024-10-01 09:00:00	CTIN 08	INV-8	12	1	0	0	0	0	0	1	0